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The Structure of Non-Hierarchical Triple System Stability Regions

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Abstract—A detailed study of the two-dimensional initial conditions region section in the planar three-body problem is performed. The initial conditions for the three well-known stable periodic orbits (the Schubart's orbit, the Broucke's orbit and the eight-like orbit) belong to this section. Continuous stability regions (for the fixed integration interval) generated by these periodic orbits are found. Zones of the quick stability violation are outlined. The analysis of some concrete trajectories coming from various stability regions is performed. In particular, trajectories possessing varying number of "eights" formed by moving triple system components are discovered. Orbits with librations are also found. The new periodic orbit originated from the zone siding with the Schubart's orbit region is discovered. This orbit has reversibility points (each of the outer bodies possess a reversibility point) and two points of close double approach of the central body to each of the outer bodies. The influence of the numerical integration accuracy on the results is studied. The stability regions structure is preserved during calculations with different values of the precision parameter, numerical integration methods and regularization algorithms of the equations of motion.

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1. INTRODUCTION

The fraction of triple systems among stars and galaxies is rather high. Approximately 10% of stars in the Solar neighborhood are members of triple systems (see [1, 2]). The fraction of triple systems among young stars in SFRs (star formation regions) is sufficiently higher than that one in the Galaxy field (see [3]). According to modern studies the prevailing channel of star formation is that one when the stars are formed as binary or triple system members [4]. Decay of non-hierarchical small stellar groups and stellar clusters can result in the formation of triple systems (see [5, 6]).

Real and model triple stars can be either dynamically stable or unstable. When a triple is unstable, it will decay into a binary and a single star.

The study of motion stability in dynamical systems is of great interest (see monograph [7] and references therein). One of the classical problems of celestial mechanics, where the stability study is of great importance, is the well-known three-body problem (see books [8, 9]). Study of hierarchical system stability plays important role in this problem. Several Hill-type and Lagrange-type stability criteria are known. Reviews of well-known stability criteria can be found in [9, 10].

The more complicated situation takes place for the non-hierarchical systems, when distances between the bodies are comparable. Several stable periodic orbits were found in such a kind of systems [11–13]. Stability criteria, that are applicable to the hierarchical systems, are not valid in this case. Therefore stability analysis is usually performed either numerically (see, e.g., [14]) or in linear approximation (see, e.g., [15]). According to KAM-theory stable periodic orbits are to be surrounded by the sets of orbits with bounded motions, at the same time the measures of these sets are not equal to null (see, e.g., [7]). These sets can possess very complicated structure, that could not be described at present analytically. The only constructive method to study this structure is numerical simulations of the great amount of the triple systems, having initial conditions in vicinity of the periodic trajectory under consideration.

In this paper numerical simulations of the triple system dynamics are used to find stability regions, generated by several stable periodic orbits, and to study the character of motions in this regions.

2. PROBLEM FORMULATION

Three well-known stable periodic orbits in general planar equal-mass three-body problem (see [11, 12,

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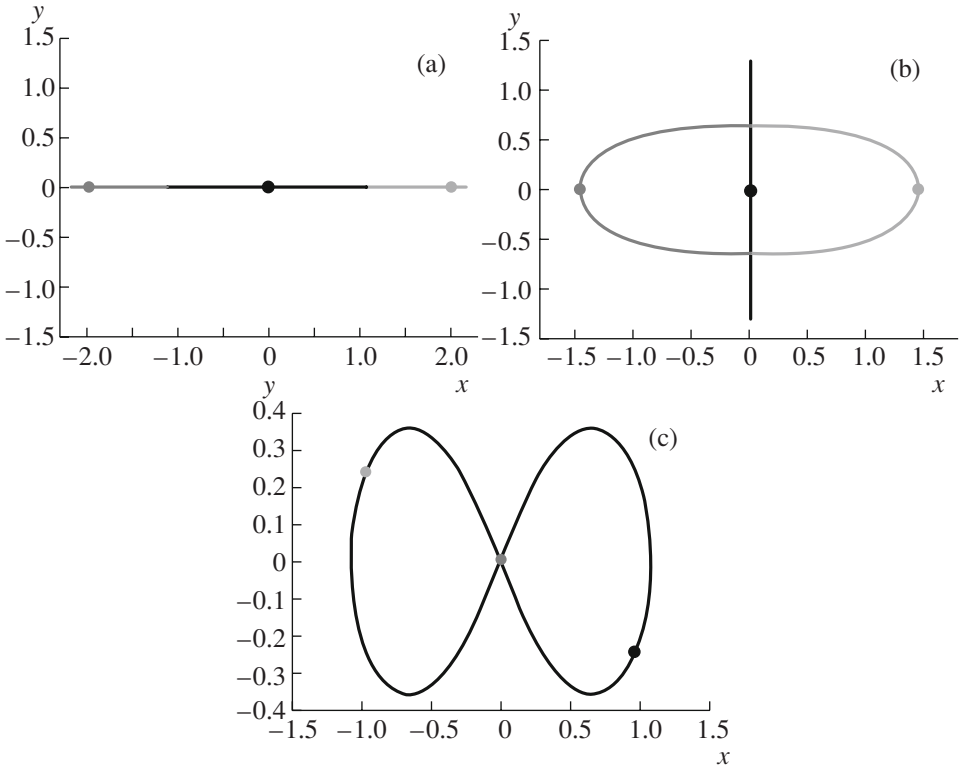


Fig. 1. Three stable periodic orbits in planar equal-mass three-body problem: (a) Schubart orbit [11]; (b) Broucke orbit [12]; (c) "eight-like" orbit [13, 16].

16]) possess one important common property (see Fig. 1). One of the bodies passes through the center-of-mass of the triple system situated between two other bodies. At this moment all three bodies are located on the straight line, having the angle between velocity vector of the central body and this straight line to be equal to zero for the Schubart orbit, $\varphi = 90°$ for the Broucke orbit, and $\varphi = 56.9°$ for the eight-like orbit.

Let us consider the family of trajectories having such kind of passages. We choose the initial conditions at the time of a passage. Taking into account center-of-mass integrals and the symmetry of the problem, we can determine initial coordinates and velocities of all bodies using only two free parameters. We take these parameters to be the angle φ and virial ratio k of the triple system, which is calculated as the ratio of the kinetic and potential energy: $k = \dfrac{T_0}{|U_0|}$. Taking into account the symmetry of the problem under consideration we have following initial condition domains: $k \in (0, 1)$ and $\varphi \in (0, \dfrac{\pi}{2})$. Let us note, that values $k = 0$ and $k = 1$ correspond to triple collisions; $\varphi = 0$ corresponds to the rectilinear problem; $\varphi = \dfrac{\pi}{2}$ corresponds to the isosceles problem. We do not study these four cases in this paper. Two latter cases are studied in other papers (see, e.g., [17, 18]).

We scan the bounded initial condition domain $k \in (0, 1)$; $\varphi \in (0, \dfrac{\pi}{2})$ with steps $\Delta k = 0.001$ and $\Delta\varphi = 0.001$. For each point in this domain numerical integration of the regularized equations of motion is performed using the code TRIPLE (author of this code is Sverre Aarseth, see in book [19]). Bulirsch-Stoer integrator [20] and Aarseth-Zare regularization [21] are applied in this code.

The integration of all trajectories under study is performed until one of the termination conditions is not satisfied:

(1) Integration time $t > 10\,000\tau$ (τ is the average time the component needs to cross a triple system). This time can be calculated using following formula:

$$\tau = \frac{G^{\frac{1}{2}}(\sum_{i=1}^{3} m_i)^{\frac{5}{2}}}{(2|E|)^{\frac{3}{2}}}, \qquad (1)$$

here m_i are masses of bodies, E is total energy of triple system, the gravitational constant G is supposed to be equal to 1).

(2) Maximum distance $r_{\max}$ between the bodies is greater than $5d$, where d is the average triple system

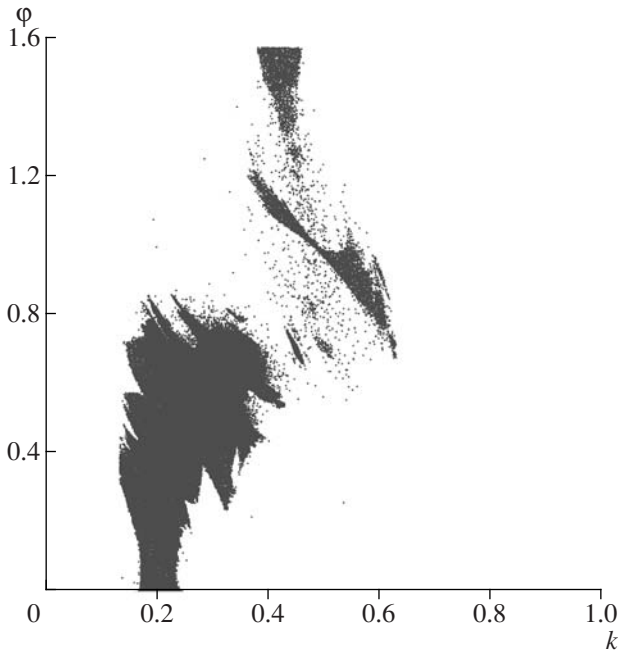


Fig. 2. Stability regions on the diagram (k, φ) at $t > 10\,000\tau$.

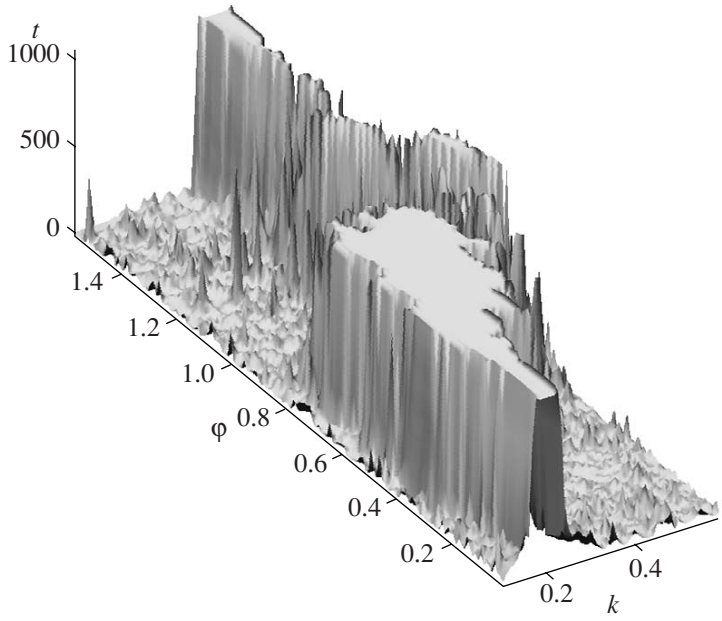


Fig. 3. Dependence of life-time on initial data (k, φ).

size, that can be calculated using following formula

$$d = \frac{G\sum_{i<j} m_i m_j}{|E|}. \qquad (2)$$

When the condition $r_{\max} > 5d$ is satisfied, then the triple system enters into ejection-state. The further evolution of such triple system will result in its decay (see, e.g., [22]). When this condition is not satisfied during a long time (e.g., $t = 10\,000\tau$) for a triple system, we can suppose that this system is stable according to Lagrange, i.e., motions in such system are bounded at least during integration time. We can not draw any conclusion about the character of motions in this system at an infinite time interval. On the other side, according to the KAM-theory, there should exist

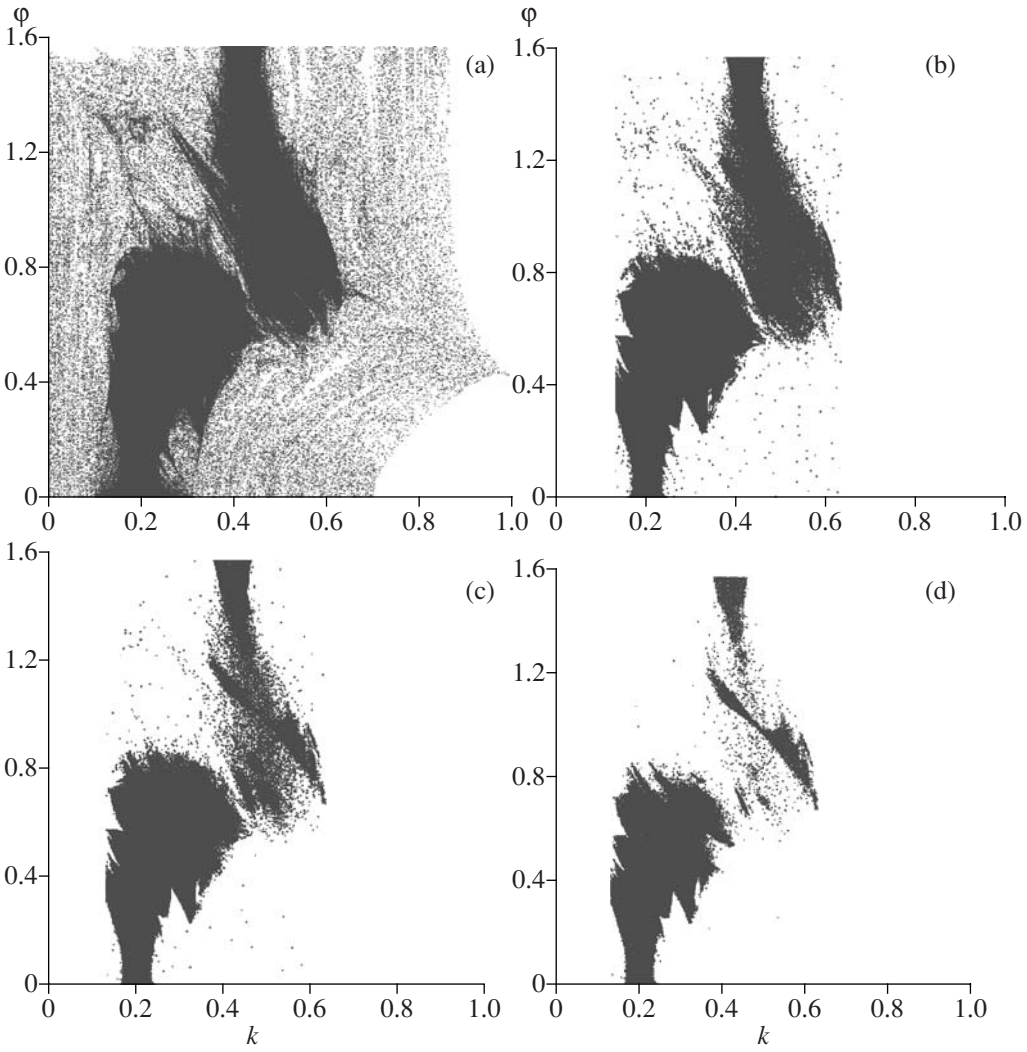


Fig. 4. Dependence of stability regions on the maximum integration interval t: (a) $t = 100\tau$; (b) $t = 500\tau$; (c) $t = 1000\tau$; (d) $t = 5000\tau$.

systems with bounded motions at an infinite time interval in vicinities of stable periodic orbits.

3. RESULTS

3.1. General Properties

General results of the initial conditions domain $k \in (0, 1)$; $\varphi \in (0, \frac{\pi}{2})$ scanning are shown in the Figs. 2, 3. One can mention continuous regions and separated points where $t > 10\,000\tau$. The largest three continuous regions are generated by three stable periodic orbits (Schubart-orbit, Broucke-orbit and eight-like orbit). These three orbits have following positions on the initial conditions plane (k, φ):

(1) Schubart-orbit—$k = 0.206$, $\varphi = 0°$;

(2) eight-like orbit—$k = 0.485$, $\varphi = 56.9°$;

(3) Broucke-orbit—$k = 0.418$, $\varphi = 90°$.

The largest region is generated by the Schubart-periodic orbit. Regions generated by Broucke- and eight-like orbits are smaller. All three regions are isolated on the initial conditions plane (k, φ). One can also found on the figures several "islands" of continuity that are smaller than three regions mentioned above. Some of these islands are located near large stability regions, some of them are placed between large stability areas.

Let us study stability region structure changes depending on the system living-time t (Fig. 4). One can see that when the living-time t grows, regions of continuity become smaller and the "blowing-up" of stability points occurs. This results in the compactness of stability regions and in the isolation growth of these regions. We can see in Fig. 4a continuous regions corresponding to fast stability violation at large k and at small k and φ near to 90°.

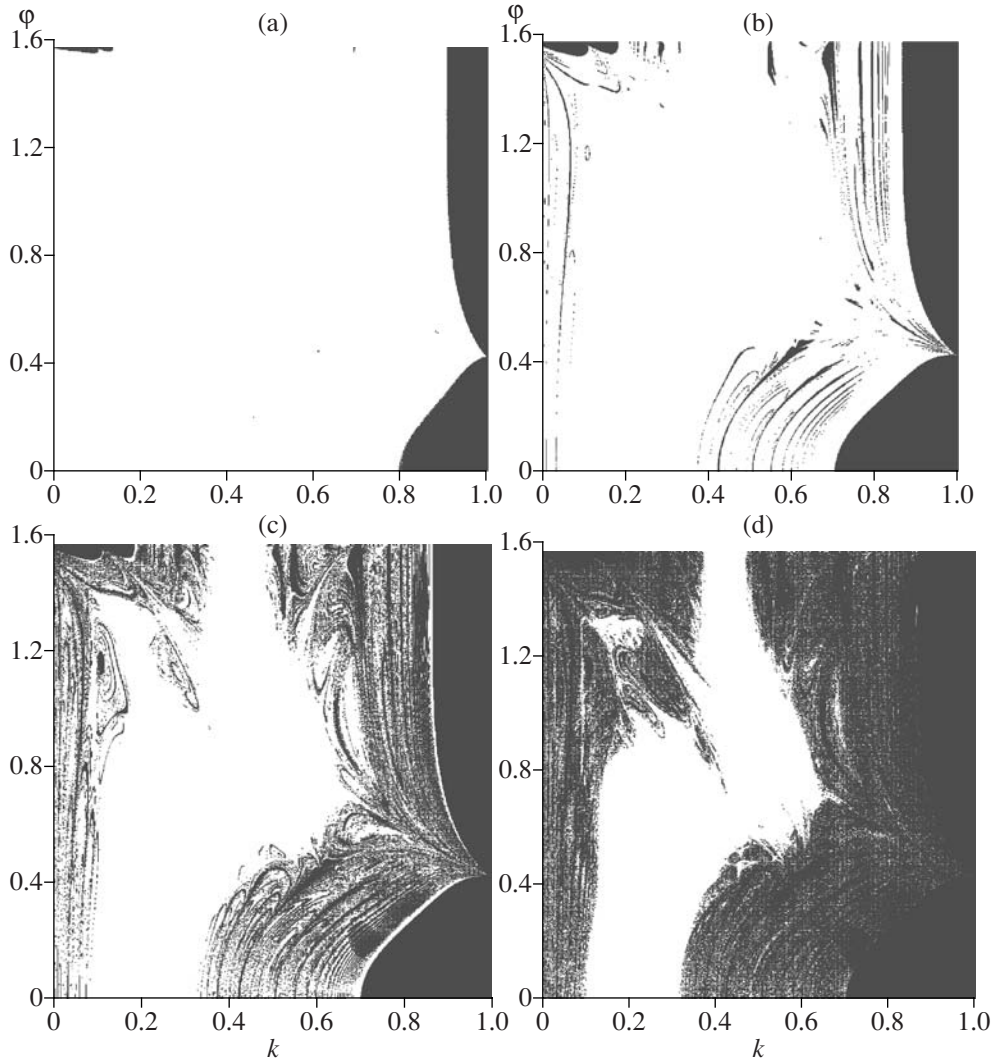


Fig. 5. Regions of stability violation on a time interval shorter than t: (a) $t = 2\tau$; (b) $t = 5\tau$; (c) $t = 10\tau$; (d) $t = 20\tau$.

It is of interest to study the structure of initial condition regions that correspond to short-living-trajectories (Fig. 5). When the integration time is small ($t < 2\tau$) one can see continuous regions of a short stability violation time-scale, where $r_{\max} > 5d$. When the living-time t grows, these regions become larger (see Fig. 5b). One can also mention elongated (frequently curved) structures where the stability is violated on a short time-scale. These structures are frequently elongated along the boundaries of the continuous regions. In continuous regions one can point out several separate points that split regions and at the same time "attract" elongated structures. When $t = 10\tau$ (Fig. 5c) there are thin strips (gaps) along the continuous region boundaries that correspond to the systems where the motions are bounded during a long time.

3.2. Trajectory Analysis

All trajectories studied one can divide into four basic types:

(1) Trajectories, where the orbit of one body relative to the center-of-mass of the remaining binary quickly becomes hyperbolic.

(2) Trajectories, where one of the bodies escapes the system after a long-term evolution, when some interactions between the bodies take place.

(3) Trajectories, where the motions are bounded at least up to chosen integration time $t = 10\,000\tau$.

(4) Periodic orbits.

Trajectories of the first type correspond to initial conditions situated in the regions of quick hierarchy violation (see Fig. 5a). At the begin of the evolution of such trajectories triple close approach takes place, that results in the escape of one body (Fig. 6a). Initial

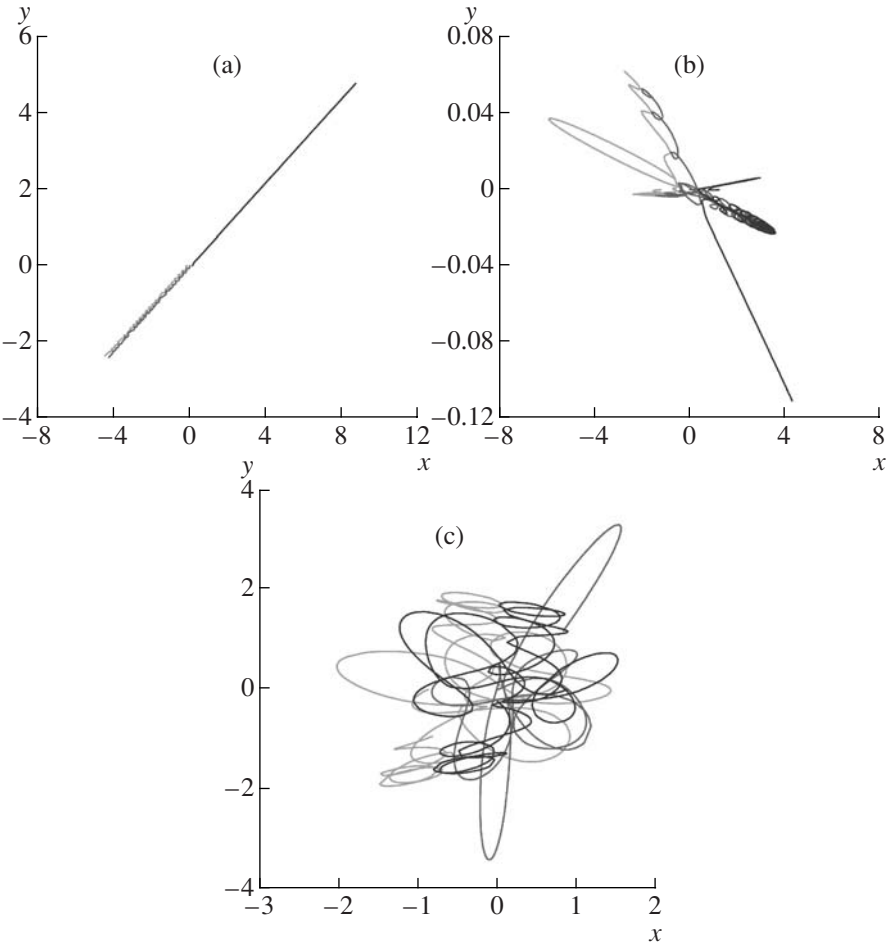


Fig. 6. Examples of unstable trajectories: (a) trajectory with fast escape of one body after triple close encounter ($k = 0.95, \varphi = 0.1$); (b) trajectory with escape after series of interactions between bodies ($k = 0.51, \varphi = 0.001$); (c) unstable long-living trajectory ($k = 0.65, \varphi = 0.6$).

conditions (k, φ) of such triple systems are located in the continuous region of quick stability violation joining to the straight line $k = 1$ (the right part of the initial conditions domain).

Trajectories of the second type have initial conditions (k, φ) outside stability regions and outside regions of quick hierarchy violation. An example of the second type trajectory is shown in the Fig. 6b. Elongated structures, that correspond to short-living trajectories (see Fig. 5c) are situated inside these initial condition regions. An example of this trajectory type is shown in the Fig. 6b. Among unstable triple systems, one can point out trajectories with long living-time and violation of stability after a big number of interactions between bodies. An example of such trajectory is shown in Fig. 6c.

Trajectories with bounded motions join to stable periodic orbits. Main features of generating stable periodic orbit trajectories (see Fig. 1) are reproduced in trajectories of this type. One can found in the Fig. 7 trajectories with initial conditions located in vicinity of Schubart, eight-like and Broucke periodic orbits. Figure 7a demonstrates that trajectories fill a bounded area during evolution and do not cross the boundary of this area.

Detailed study of some trajectories, that have initial conditions inside stability regions, shows that there exist more complicated periodic orbits inside these regions. An example of such complicated periodic orbit generated by the Schubart periodic orbit is shown in the Fig. 8. Let us note, that the orbit of the central body forms several "figures-of-eight," that are elongated in direction of triple system symmetry axis (axis of symmetry is located in vicinity of X-axis). External envelope of the central body orbit also forms an eight-like figure having intersections in zones, where the central body approaches to other bodies. At the same time peripheral bodies move along trajectories, having eight-like form, that are oriented approximately orthogonal to the symmetry axis. The number

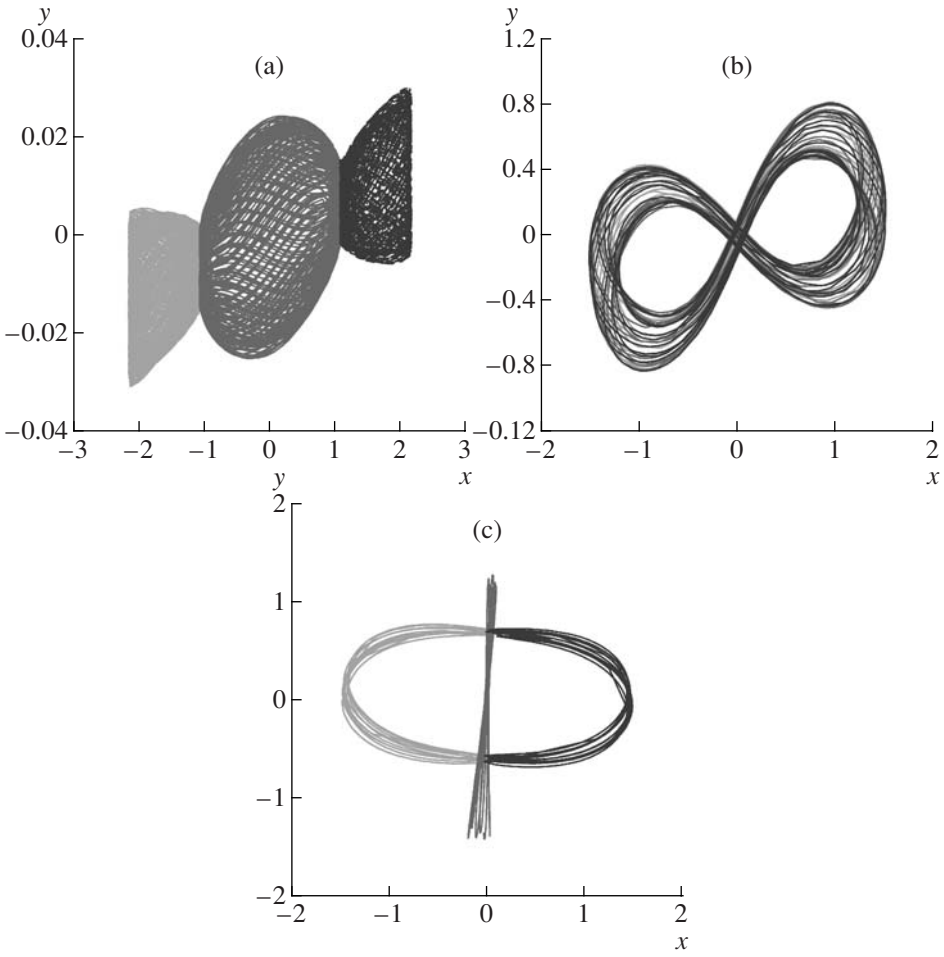


Fig. 7. Trajectories with bounded motions in vicinity of periodic orbits: (a) Schubart orbit ($k = 0.2, \varphi = 0.02$); (b) "eight-like" orbit ($k = 0.55, \varphi = 0.9$); (c) Broucke orbit ($k = 0.41, \varphi = 1.55$).

of "eights" formed by the central body trajectory can be various. In some cases these "eights" can engage each other, in other cases they are isolated.

A new periodic orbit, located at the special place on the plane $(k, \varphi) \approx (0.33, 0.66)$, is discovered. Trajectories of bodies in vicinity of this orbit are shown in the Fig. 9a. This orbit has a form of "disrupted" Schubart orbit with two return points for peripheral bodies (one point for each body) and with two double collision points of the central body with peripheral ones. Initial conditions of this triple system are located in the region, joining to the upper boundary of the stability region generated by the Schubart orbit. Small variations of initial conditions lead to librations and to appearance of the tubes, filled by trajectory coils (Fig. 9b).

This trajectory is a transient one between trajectories of the Schubart-type and "figure-of-eight-type." Larger variations of initial conditions in direction from Schubart orbit's region to eight-like orbit's region lead to appearance of transient orbits, possessing simultaneously Schubart orbit's properties (the central body approaches each of the peripheral bodies) and the ones of eight-like orbit (trajectories of all bodies are "eights" oriented in the same direction). An example of such transient orbit is shown in the Fig. 10a. Orbit of each body forms a "disrupted eight," which is turned up twice.

The second transient zone is located between orbits generated by eight-like orbit and Broucke orbit. Trajectories in this region are similar to both orbits (Fig. 10b) and have some properties of Schubart orbit. Let us note, that all the bodies form "eight-like" trajectories oriented in the same direction. The difference between these orbits and the Schubart orbit is the orientation of the eight-like trajectories: the orbits of peripheral bodies are orthogonal to the orbit of the central body in the Schubart orbit; in these orbits trajectories of all three bodies are oriented in the same direction.

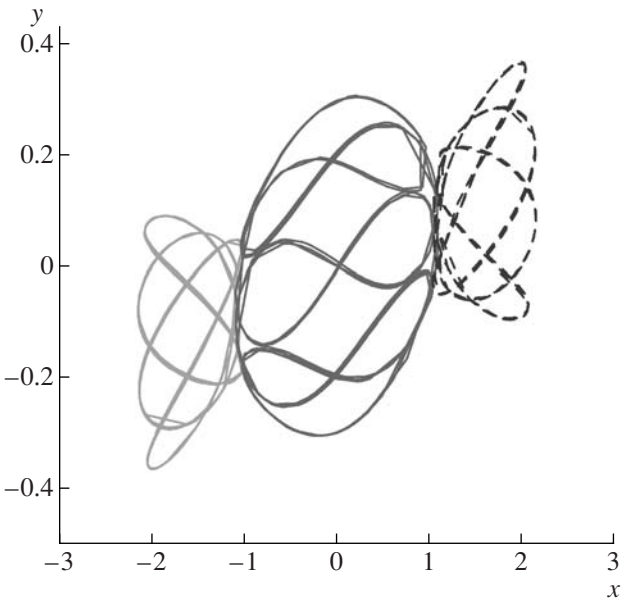


Fig. 8. An example of trajectories in vicinity of more complicated periodic orbit generated by Schubart orbit ($k = 0.18, \varphi = 0.7$).

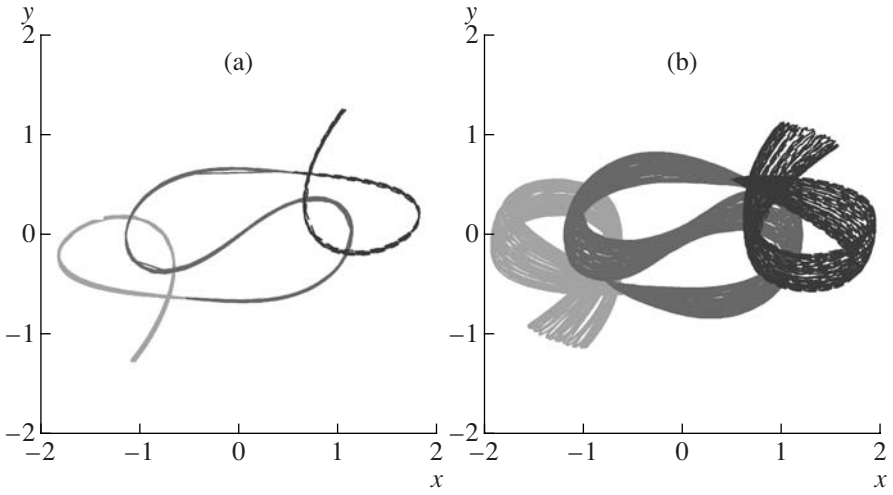


Fig. 9. Trajectories in vicinity of new periodic orbit: (a) ($k = 0.33, \varphi = 0.66$); (b) ($k = 0.275, \varphi = 0.75$) at the time interval $t = 100\tau$.

3.3. Accuracy Problem

Since numerical integration of long-living orbits is performed on a long time interval, we have a problem of reliability of our results. To check out our results we scan initial condition region with steps $\Delta k = 0.001$ and $\Delta\varphi = 0.001$, and integrate each trajectory threefold, using following tolerance parameters $\epsilon = 10^{-13}$, 10^{-14} and 10^{-15}. The results of comparison are shown in Fig. 11. General structure of stability regions remains unchanged by variation of ϵ parameter. Nevertheless there exist several differences. Stability region in vicinity of Schubart orbit is discontinuous, when parameters are chosen as follows $\epsilon = 10^{-13}$ and 10^{-14}. When parameter ϵ is equal to 10^{-15}, the region remains continuous. Appearance of these gaps is likely to be connected with error accumulation on a long integration interval.

We also use another regularization algorithm (chain regularization [23]) and another integrator (Runge-Kutta method of 8th order) to verify our

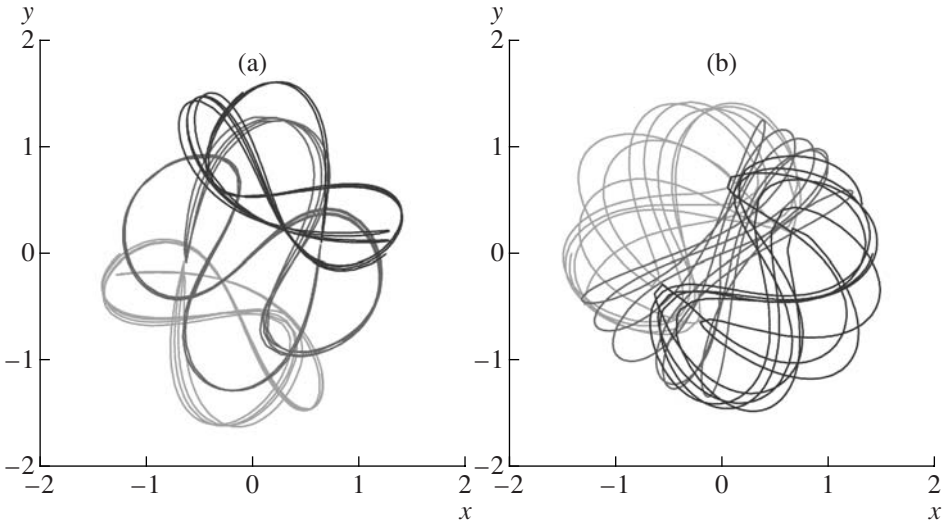


Fig. 10. Transient trajectories: (a) between region of new periodic orbit and that one of "eight-like" orbit ($k = 0.5, \varphi = 0.7$); (b) between region of "eight-like" orbit and that one of Broucke orbit ($k = 0.435, \varphi = 1.28$).

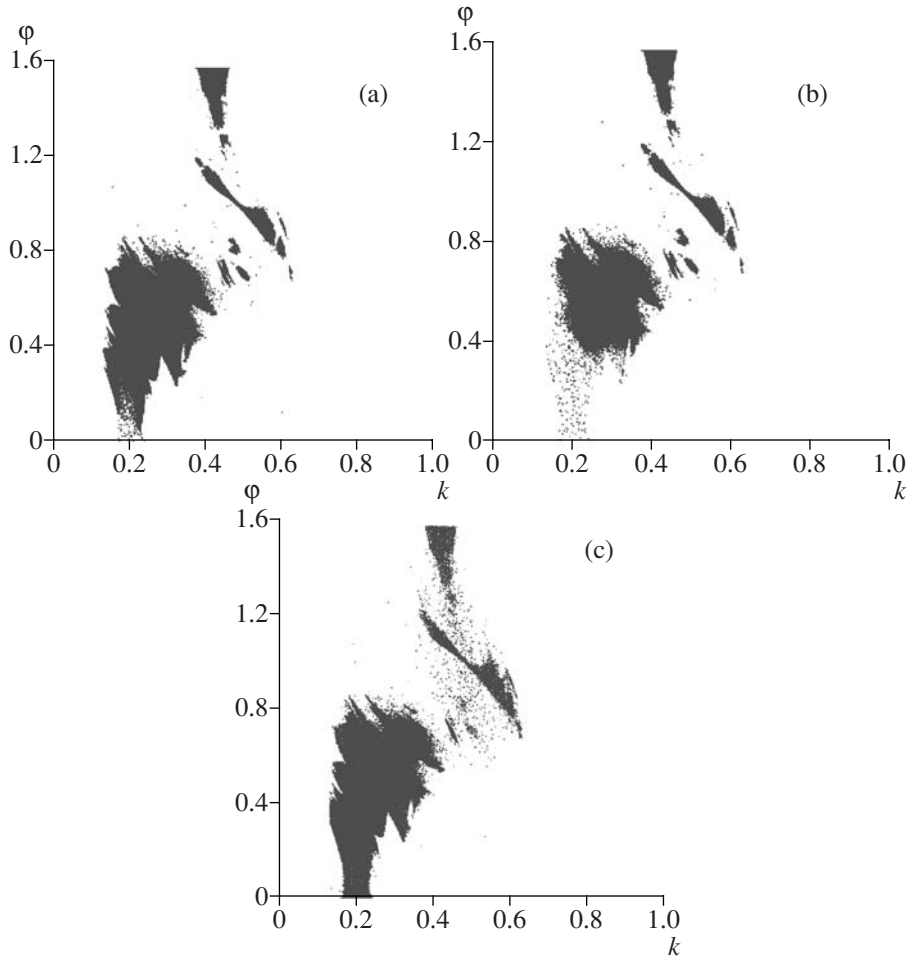


Fig. 11. Stability regions on diagram (k, φ) for $t > 10\,000\tau$ at three tolerance parameters: (a) $\epsilon = 10^{-13}$; (b) $\epsilon = 10^{-14}$; (c) $\epsilon = 10^{-15}$.

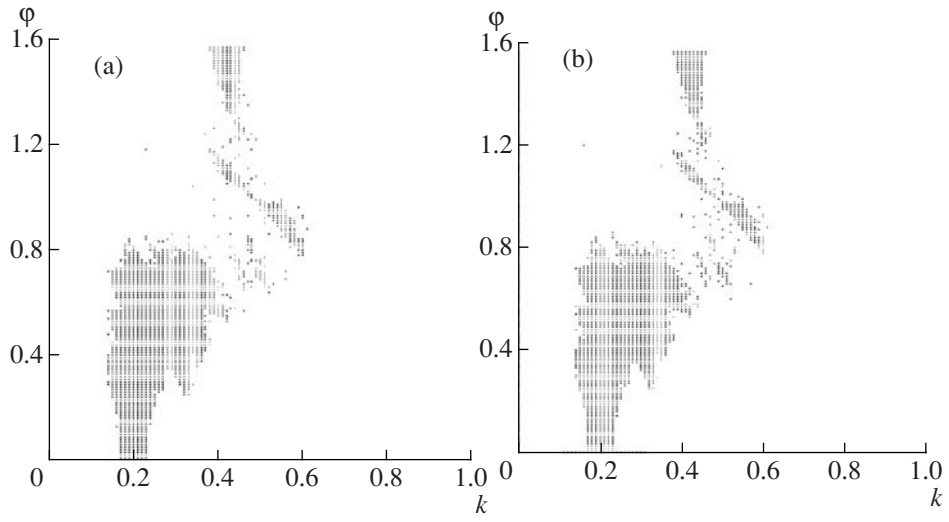


Fig. 12. Stability regions at $t > 1000\tau$ using two integration methods and regularization algorithms: (a) Bulirsch–Stoer integrator and Aarseth–Zare regularization; (b) Runge–Kutta 8th order integrator and Mikkola–Aarseth chain regularization.

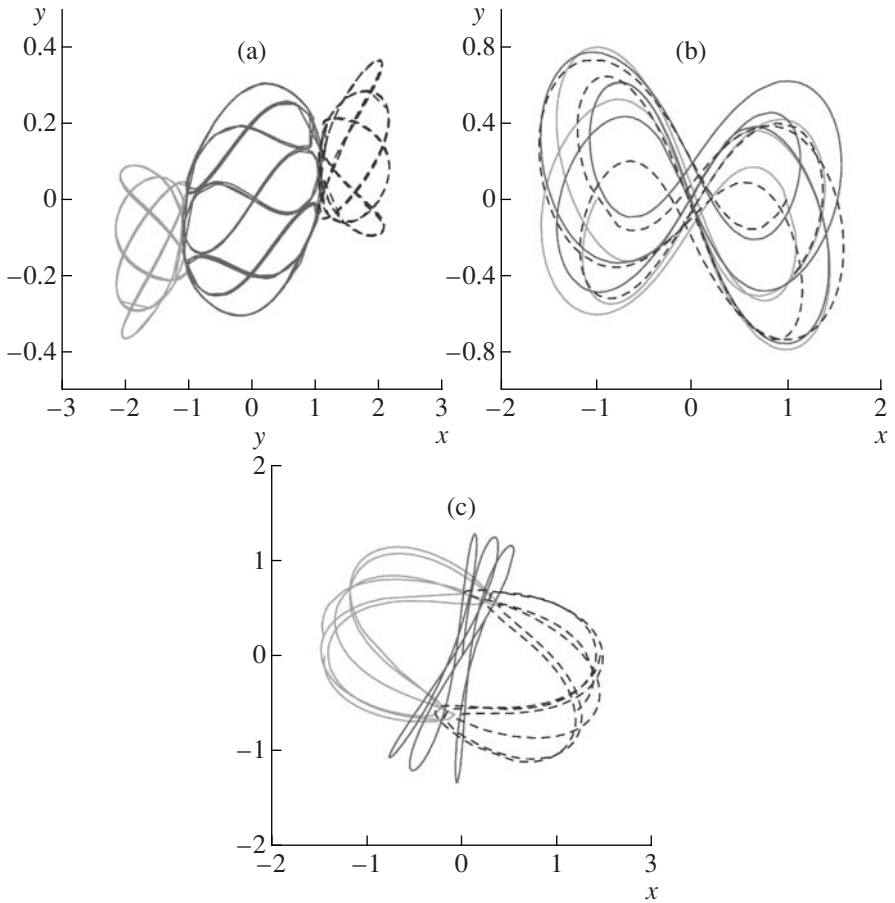


Fig. 13. Trajectories in vicinity of three periodic orbits: (a) Schubart orbit; (b) "eight-like" orbit; (c) Broucke orbit.

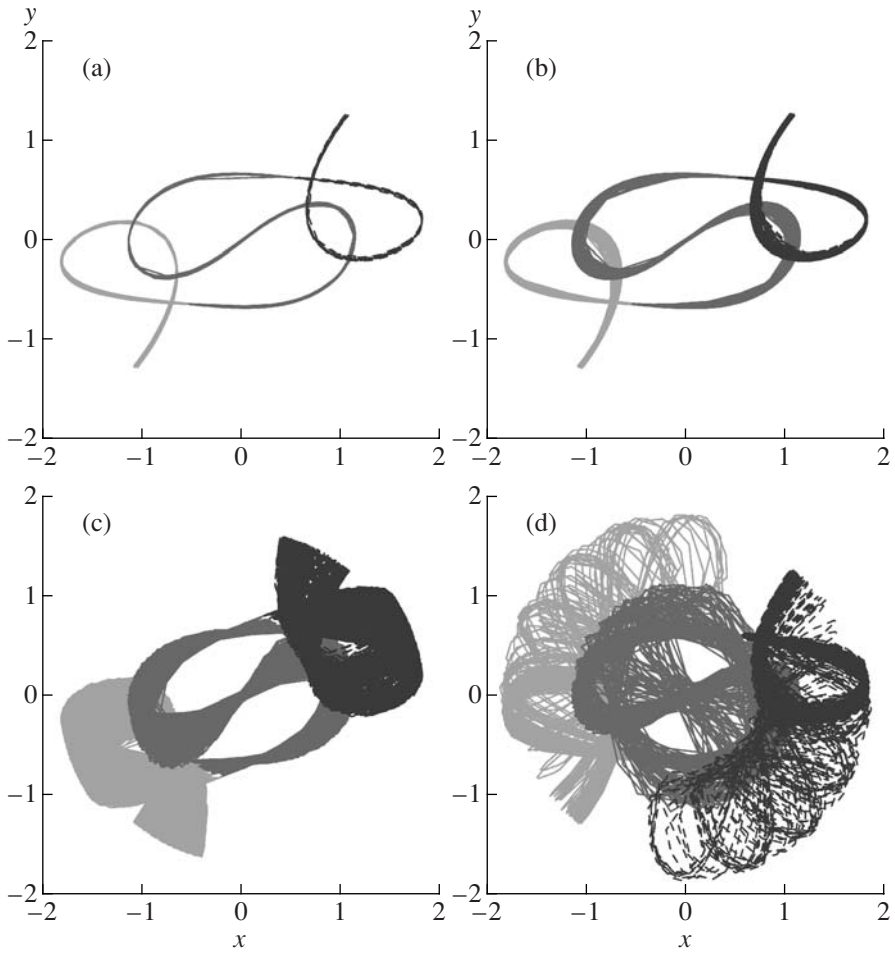


Fig. 14. Influence of the tolerance parameter on the near-periodic trajectory ($k = 0.33, \varphi = 0.66$): (a) $\epsilon = 1.5 \times 10^{-16}$ ($t = 10\tau$); (b) $\epsilon = 5 \times 10^{-16}$ ($t = 500\tau$); (c) $\epsilon = 10^{-4}$ ($t = 500\tau$); (d) $\epsilon = 10^{-2}$ ($t = 300\tau$).

results. We perform integration on a time interval $t = 1000\tau$ with steps $\Delta k = \Delta\varphi = 0.01$ using both methods. Results are shown in the Fig. 12. The results are in agreement: shapes of stability regions are similar. Hence we can conclude that our results are reliable at least up to $t \sim 1000\tau$.

It is of interest to perform extensive verification using other integration methods, e.g. symplectic methods.

4. DISCUSSION

A detailed study of two-dimensional cross-section of initial conditions region in planar three-body problem is performed. The feature of this cross-section is that fact, that initial conditions of three well-known periodic orbits (Schubart orbit [11], Broucke orbit [12] and eight-like orbit [13, 16]) belong to it. Continuous stability regions (at the fixed value of integration time), generated by these periodic orbits, are found in this cross-section.

Let us note, that stability region, generated by eight-like orbit is located between stability regions, generated by Schubart and Broucke periodic orbits. Trajectories with initial conditions located inside these stability regions may be described as "eights," having different orientation (see Fig. 13).

Stability regions have complicated structure:

(1) There exist "islands" in vicinity of three largest stability regions, that probably correspond to other periodic orbits; these "islands" are separated by "straits" of unstable orbit initial conditions.

(2) Stability region boundaries have complicated (probably fractal) structure.

(3) There exist zones of long-living unstable orbits between stability regions.

Besides stability regions, zones of quick stability violation are found. Most of them are concentrated in vicinity of straight line $k = 1$. There exist elongated structures in these zones, that are located at the boundaries of these zones (see Fig. 5). Let us note

that these structures are separated by a line, joining two "critical points" $(k, \varphi) \approx (1, \frac{\pi}{8})$ and $(k, \varphi) \approx (0.1, \frac{\pi}{2})$. It is of interest to study trajectories located at various structures, where quick stability violation occures.

Besides a review of initial condition domain (k, φ) the analysis of some concrete trajectories originating from various stability regions is performed. In the stability region near the Schubart periodic solution trajectories having different number of "eights" (e.g., Fig. 8) are found. Orbits with librations are also found (see, Figs. 7a and 9b).

A new periodic orbit is found in the region joining to the Schubart periodic solution (see Fig. 9a). This orbit has two return points (one point for each peripheral body) and two points of double approaches of the central and peripheral bodies. Several transient trajectories between three stable orbits are also found.

The influence of accuracy of calculations on the results is studied. General stability region structure remains unchanged, when different tolerance parameters, integration methods or regularization algorithms are used (see Figs. 11 and 12).

The influence of tolerance parameter ϵ on the stable trajectory behaviour is studied (see Fig. 14). Variation of ϵ parameter in a wide interval results in a similar shape of trajectories; the tubes may become thicker, provided the tolerance parameter is large and integration is performed on a long time interval (forking of return points is also possible). When an inadequate tolerance parameter is chosen orbit precession occurs. Let us note, that when $t > 500\tau$, trajectory on the Fig. 14 fills a ring. Integration with lower precision leads to higher rate of the fictive trajectory evolution.

The study of this initial conditions domain can be continued in following directions:

(1) Study of stability region evolution with the growth of integration time;

(2) Detailed analysis of trajectory features, originated from different zones of stable motions and quick stability violation;

(3) Search for a dependency between accuracy of integration and trajectory evolution.

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